System(3X±1)/4 Framework for the Collatz (3x+1) Problem : A Definitive Proof of the Collatz Conjecture

© Dr.(Prof.) Keshava Prasad Halemane,
Professor - retired (2017JAN31) from
Department of Mathematical And Computational Sciences,
National Institute of Technology Karnataka Surathkal,
Srinivasnagar, Mangaluru - 575025, India.
https://orcid.org/0000-0003-3483-3521

Independent Researcher @ SASHESHA
(no affiliations; no funding; no data used; no conflict of interests)
https://www.linkedin.com/in/keshavaprasadahalemane/
k.prasad.h@gmail.com

Abstract

We design a mathematical framework based on System(3X±1)/4; defined similar to the Collatz (3x+1) system, with slightly modified operations. On similar lines, we define the CTUHSK(3x+1)/4 system, and show that to be exactly the same as the Collatz (3x+1) system. We establish the convergence of System(3X±1)/4 to its trivial-cycle {(1⇐(2)⇐4)}; using the persistent local characteristic property of 'odd-to-odd monotonicity' in its defining function. Therefore, the transitive closure of the subset {1,5,3} under the inverse mapping of System(3X±1)/4; represented by the nodes of the system arborescence; is the entire set of positive integers. We present a meticulously designed reversible (invertible) transformation (reduction); between System(3X±1)/4 and the CTUHSK(3x+1)/4 system; based on pruning-

&-joining operations on the system arborescence; which does not affect the binary(inward)arborescence structure and the connectedness property of either of these systems. We establish that the transitive-closure of the subset {1,5,3} under the inverse mapping of CTUHSK(3x+1)/4 system is exactly the same as the transitive-closure of that subset under the inverse mapping of System(3X±1)/4; which has been shown to be the entire set of all natural numbers. This establishes the convergence of the CTUHSK(3x+1)/4 system and therefore the Collatz (3x+1) system, to the trivial-cycle {(1⇐2⇐4)}; thus providing a definitive proof of the Collatz Conjecture.

This paper also presents Halemane-Conjecture on the Collatz sequence, that the upper bound on the number of odd (3x+1) operations required to reach its trivial-cycle {(1⇐2⇐4)} starting from any given positive integer and moving along the Collatz sequence, is the given number itself; the triad {(31⇐41⇐27)} being an exceptional limiting case.

Keywords: System(3X±1)/4; CTUHSK(3x+1)/4 system;
Halemane's Conjecture on the Collatz (3x+1) system;
Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Sequence;
Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Conjecture;
binary(inward)arborescence;

AMS MSC Mathematics Subject Classification: 11B83; 11Y55; 40A99; 93A99.

1. Introduction

The Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Conjecture, simply referred to as the *Collatz Conjecture* [§1] states that the Collatz (3x+1) Sequence, also referred to as the Collatz Sequence, converges to its trivial-cycle; that is, {(1⇐2⇐4)}, starting from any positive integer.

Lagarias [§2]&[§3] gives an exhaustive annotated bibliography, whereas Lagarias [§4]&[§5] gives an elaborate overview of the Collatz Problem, also referred to as the "3x+1 problem", referring to it as "The Ultimate Challenge". Guy [§6] has been somewhat pessimistic in advising researchers "Don't Try to Solve These Problems". Lagarius [§4]&[§5] refers to the statement by the most revered Mathematician and Number Theory Expert Paul Erdos, who said that - "Mathematics is not yet ready for such problems" - about the Collatz Conjecture. The Collatz (3x+1) Sequence is indeed a highly complex chaotic system, although defined or generated with one of the simplest deterministic well-defined equations, that too, just two in number. We will not get into any more of the reported literature except the problem definition; but will simply focus on a rather shockingly simple approach toward a possible resolution of this amazingly notorious long-standing unresolved problem.

This paper presents System(3X±1)/4 that is defined on similar lines as the Collatz (3x+1) system with slightly modified operations. Also, on similar lines, the CTUHSK(3x+1)/4 system is defined and is shown to be exactly the same as the Collatz (3x+1) system. The System(3X±1)/4 is shown to converge to its trivial-cycle {(1⇐(2)⇐4)}; using the local 'odd-to-odd monotonicity' property of the defining function. A reversible (invertible) transformation (reduction) has been

designed; between System(3X±1)/4 and the CTUHSK(3x+1)/4 system; based on pruning-&-joining operations on the system arborescence; is shown not to affect the binary(inward)arborescence structure and the connectedness property of either of these systems. It is shown that the transitive-closure of the subset {1,5,3} under the inverse mapping of CTUHSK(3x+1)/4 system is exactly the same as the transitive-closure of that subset under the inverse mapping of System(3X±1)/4; which is the entire set of all natural numbers; thus establishing the convergence of the CTUHSK(3x+1)/4 system to its trivial-cycle {(1⇐2⇐4)}.

2. System(3X±1)/4

System(3X±1)/4 H is defined by its immediate-successor function S(x) as follows:

For any positive integer x;
if $x \equiv 0 \pmod 4$ then, $S(x) := (x/4)$;
elseIf $x \equiv 2 \pmod 4$ then, $S(x) := (x/2)$;
elseIf $\{x \equiv 1 \pmod 4\}$ then, $S(x) := (3x+1)$;
elseIf $\{x \equiv 3 \pmod 4\}$ then, $S(x) := (3x-1)$; [Eqn.1]

Note that as per the above definition, for every positive integer x; $y := S(x)$ is the single unique immediate-successor of x. The 4x4 state transition matrix H(x, y) for System(3X±1)/4 is given in Table-1. Note that $[c]_b$ represents the modular (modulo-b) residue class c that a given positive integer belongs to.

H(x,y)	$[0]_4$	$[2]_4$	$[1]_4$	$[3]_4$
$[0]_4$	1	1	1	1
$[2]_4$	0	0	1	1
$[1]_4$	1	0	0	0
$[3]_4$	1	0	0	0

Table-1: System(3X±1)/4

In order to gain better understanding of the structure of System(3X±1)/4; we first construct the inverse function defined by the immediate-predecessor relation {P(y)} derived from the above definition; that is -

For any given positive integer y;

if $y \in \{[0]_{12}; [2]_{12}; [6]_{12}; [10]_{12}\}$; then, $\{P(y)\} \coloneqq \{(4y)\}$;

if $y \in \{[1]_4; [3]_4\}$; then, $\{P(y)\} \coloneqq \{(2y); (4y)\}$;

if $y \in \{[4]_{12}\}$; then, $\{P(y)\} \coloneqq \{(4y); (y-1)/3\}$;

if $y \in \{[8]_{12}\}$; then, $\{P(y)\} \coloneqq \{(4y); (y+1)/3\}$; [Eqn.2]

Note that for any given positive even number, $y \in \{[0]_4; [2]_4\}$; there exists exactly one *immediate-even-predecessor*, $x \coloneqq 4y$; whereas any given positive odd integer, $y \in \{[1]_4; [3]_4\}$; has exactly two *immediate-even-predecessors*, $\{x\} \coloneqq \{(2y); (4y)\}$. Also, a positive even number $y \in \{[4]_{12}\}$ has exactly one *immediate-odd-predecessor*, $x \coloneqq (y-1)/3$; and a positive even number $y \in \{[8]_{12}\}$ has exactly one *immediate-odd-predecessor*, $x \coloneqq (y+1)/3$; but, a positive even number $y \in \{[0]_{12}; [2]_{12}; [6]_{12}; [10]_{12}\}$ cannot have any *immediate-odd-predecessor*. These characteristic properties result

in a '*binary*(inward)*arborescence*' structure for the System(3X±1)/4; with its trivial-cycle {(1⇐(2)⇐4)} as a root; and every odd multiple of 3 being an 'odd-leaf-node'; that is, none of its even predecessors can have any immediate-odd-predecessor.

Table-2 gives the expanded 12x12 state transition matrix H(x, y) for System(3X±1)/4, that explicitly captures and represents the above-mentioned characteristic properties.

H(x,y)	$[0]_{12}$	$[4]_{12}$	$[8]_{12}$	$[2]_{12}$	$[6]_{12}$	$[10]_{12}$	$[1]_{12}$	$[5]_{12}$	$[9]_{12}$	$[3]_{12}$	$[7]_{12}$	$[11]_{12}$
$[0]_{12}$	1	0	0	0	1	0	0	0	1	1	0	0
$[4]_{12}$	0	1	0	0	0	1	1	0	0	0	1	0
$[8]_{12}$	0	0	1	1	0	0	0	1	0	0	0	1
$[2]_{12}$	0	0	0	0	0	0	1	0	0	0	1	0
$[6]_{12}$	0	0	0	0	0	0	0	0	1	1	0	0
$[10]_{12}$	0	0	0	0	0	0	0	1	0	0	0	1
$[1]_{12}$	0	1	0	0	0	0	0	0	0	0	0	0
$[5]_{12}$	0	1	0	0	0	0	0	0	0	0	0	0
$[9]_{12}$	0	1	0	0	0	0	0	0	0	0	0	0
$[3]_{12}$	0	0	1	0	0	0	0	0	0	0	0	0
$[7]_{12}$	0	0	1	0	0	0	0	0	0	0	0	0
$[11]_{12}$	0	0	1	0	0	0	0	0	0	0	0	0

Table-2: System(3X±1)/4 - 12x12 State Transition Matrix

3. Observations on the Structure of System(3X±1)/4 H

The binary(inward)arborescence structure of the System(3X±1)/4 H has the following characteristic properties:

(0) every node has exactly one immediate-successor.

(1) every odd node has exactly two immediate-even-predecessors.

(2) every even node has exactly one immediate-even-predecessor.

(3) every even node $x \in \{[4]_{12}; [8]_{12}\}$ has exactly one immediate-odd-predecessor; such nodes are called 'active-nodes'.

(4) no even node $x \in \{[0]_{12}; [2]_{12}; [6]_{12}; [10]_{12}\}$ has any immediate-odd-predecessor; such nodes are called 'inactive-nodes'.

(5) every odd-node $x \in [3]_6 \coloneqq \{[3]_{12}; [9]_{12}\}$;

that is, x being an odd-multiple-of-3; is an 'odd-leaf-node';

none of its even predecessors has any immediate-odd-predecessor.

4. Convergence of System(3X±1)/4

Note that every application of the (3X±1) operation produces an immediate resultant positive even numbers $x \in \{[0]_4\}$; and therefore every (3X±1) operation *must necessarily be followed* by *at least one* application of the (x/4) operation; that may even be followed by further repetitions of it; leading to an *immediate decrease* in the numerical value of the operand. Therefore, the resulting odd-to-odd subsequence is a "*monotonically strictly decreasing odd-to-odd subsequence*" – a characteristic property that we refer to as, a persistent local 'odd-to-odd monotonicity'. This immediately establishes our assertion, that –

Assertion-H-1:

The System(3X±1)/4 converges to its trivial-cycle {(1⇐(2)⇐4)}; starting with any positive integer x > 1.

This convergence property Assertion-H1 immediately implies Assertiob-H-2:

Assertion-H-2:

The transitive-closure of the subset {1,5,3}; that is, [{1,5,3}H+]; under the inverse mapping of System(3X±1)/4; represented by the nodes of the system arborescence; is the entire set of positive integers N; that is, [{1,5,3}H+] = {nodes(H)} = N.

5. Collatz (3x+1) system re-written as CTUHSK(3x+1)/4 system

The Collatz(3x+1)/2System can be defined by its immediate-successor function S(x) for the (classical/canonical) Collatz sequence that can be written as –

For any positive integer x;
if x≡0(mod2) then, S(x) ≔ (x/2);
elseIf {(x≡1(mod2)} then, S(x) ≔ (3x+1); [Eqn.3]

Note that as per [Eqn.3]; for every positive integer x there exists a single unique immediate-successor y.

Also note that the immediate-successor of any positive odd number x of the form (4m+1) in the Collatz sequence is a positive even number of the form (12m+4);

that is, a number y∈{$[0]_4$} that is always divisible by 4 *at least once*, and therefore the combined operation can be denoted by a single ((3x+1)/4) operation; which leads to a clear decrease in the numerical value for the resultant odd number.

6. Critical-Issue, Critical-(Decision)-Node, Detour-Link, Detour-Node

Observe that the immediate-successor of any positive odd number x of the form (4m-1) in the Collatz sequence is a positive even number y of the form (12m-2); that is, a number y ∈ {$[2]_4$} that is always divisible by 2 *exactly once* only. The combined operation can be denoted by a single ((3x+1)/2) operation, with its immediate resultant being a positive odd number z of the form (6m-1); with an increase in the numerical value of the operand. We designate this specific step {(12m-2)⇐(4m-1)} in the Collatz system as a critical *detour-link*. The critical detour-link is the very source for the possible ambiguity (*critical-issue*) regarding the convergence of the Collatz sequence. We designate the initial-node (4m-1) of a detour-link as its *critical-*(*decision*)*-node*; and its terminal-node (12m-2) as its *detour-node*. This *critical-issue* is addressed appropriately here in this paper.

As a first step, the Collatz (3x+1) system originally defined by [Eqn.3] can now be re-written as an equivalent CTUHSK(3x+1)/4 system; with a special single application of a (x/2) operation in such specific cases; whenever the operand is a positive even number of the form (12m-2) arising as the immediate resultant of the application of the (3x+1) operation on a positive odd number of the form (4m-1).

The Collatz (3x+1) system; now re-written & renamed as its equivalent CTUHSK(3x+1)/4 system; is defined by its immediate-successor function, as -

For any positive integer x;

if $x \equiv 0 (\text{mod} 4)$ then, $S(x) := (x/4)$;

elseIf $x \equiv 2 (\text{mod} 4)$ then, $S(x) := (x/2)$;

elseIf $\{(x \equiv 1 (\text{mod} 4)) \vee (x \equiv 3 (\text{mod} 4))\}$ then, $S(x) := (3x+1)$; [Eqn.4]

Assertion-C-1:

The CTUHSK(3x+1)/4 system C is exactly identical to the Collatz (3x+1) system; by its very definition.

Note that as per the above definition of CTUHSK(3x+1)/4 system; every positive integer x; has its single unique immediate-successor, $y := S(x)$. The 4x4 state transition matrix C(x, y) for CTUHSK(3x+1)/4 system C is given in Table-3.

C(x,y)	$[0]_4$	$[2]_4$	$[1]_4$	$[3]_4$
$[0]_4$	1	1	1	1
$[2]_4$	0	0	1	1
$[1]_4$	1	0	0	0
$[3]_4$	0	1	0	0

Table-3: CTUHSK(3x+1)/4 system

In order to gain better understanding of the structure of CTUHSK(3x+1)/4 system; we construct the inverse function, defined by the immediate-predecessor relation {P(y)}; derived from the above definition; that is -

For any positive integer y;
if y ∈ {$[0]_{12}$; $[2]_{12}$; $[6]_{12}$; $[8]_{12}$}; then, {P(y)} ≔ {(4y)};
if y ∈ {$[1]_4$; $[3]_4$}; then, {P(y)} ≔ {(2y); (4y)};
if y ∈ {$[4]_{12}$; $[10]_{12}$}; then, {P(y)} ≔ {(4y); (y-1)/3}; [Eqn.5]

Note that for any given positive even number, y∈ {$[0]_4$; $[2]_4$}; there exists exactly one *immediate-even-predecessor*, x≔ 4y; whereas, any given positive odd integer, y∈ {$[1]_4$; $[3]_4$}; has exactly two *immediate-even-predecessors*, {x}≔ {(2y); (4y)}. Also, a positive even number y∈ {$[4]_{12}$; $[10]_{12}$} has exactly one *immediate-odd-predecessor*, x≔ (y-1)/3; but, a positive even number y∈ {$[0]_{12}$; $[2]_{12}$; $[6]_{12}$; $[8]_{12}$} cannot have any *immediate-odd-predecessor*. These characteristic properties result in a '*binary*(inward)*arborescence*' structure for the CTUHSK(3x+1)/4 system; with its trivial-cycle {(1⇐2⇐4)} as a root; and every odd multiple of 3 being an 'odd-leaf-node'; that is, none of its even predecessors can have any immediate-odd-predecessor.

Table-4 gives the expanded 12x12 state transition matrix C(x, y) for CTUHSK(3x+1)/4 system, that explicitly captures and represents the above-mentioned characteristic properties.

The highlighted entries in Table-3 & Table-4 indicate the differences in the CTUHSK(3x+1)/4 system as compared with the corresponding entries in System(3X±1)/4 presented in Table-1 & Table-2.

C(x,y)	$[0]_{12}$	$[4]_{12}$	$[8]_{12}$	$[2]_{12}$	$[6]_{12}$	$[10]_{12}$	$[1]_{12}$	$[5]_{12}$	$[9]_{12}$	$[3]_{12}$	$[7]_{12}$	$[11]_{12}$
$[0]_{12}$	1	0	0	0	1	0	0	0	1	1	0	0
$[4]_{12}$	0	1	0	0	0	1	1	0	0	0	1	0
$[8]_{12}$	0	0	1	1	0	0	0	1	0	0	0	1
$[2]_{12}$	0	0	0	0	0	0	1	0	0	0	1	0
$[6]_{12}$	0	0	0	0	0	0	0	0	1	1	0	0
$[10]_{12}$	0	0	0	0	0	0	0	1	0	0	0	1
$[1]_{12}$	0	1	0	0	0	0	0	0	0	0	0	0
$[5]_{12}$	0	1	0	0	0	0	0	0	0	0	0	0
$[9]_{12}$	0	1	0	0	0	0	0	0	0	0	0	0
$[3]_{12}$	0	0	0	0	0	1	0	0	0	0	0	0
$[7]_{12}$	0	0	0	0	0	1	0	0	0	0	0	0
$[11]_{12}$	0	0	0	0	0	1	0	0	0	0	0	0

Table-4: CTUHSK(3x+1)/4 system 12x12 State Transition Matrix

7. Observations on the Structure of CTUHSK(3x+1)/4 system C

The binary(inward)arborescence structure of the CTUHSK(3x+1)/4 system C has the following characteristic properties:

(0) every node has exactly one immediate-successor.

(1) every odd node has exactly two immediate-even-predecessors.

(2) every even node has exactly one immediate-even-predecessor.

(3) every even node x $\in \{[4]_{12}; [10]_{12}\}$ has exactly one immediate-odd-predecessor;

such nodes are called 'active-nodes'.

(4) no even node x ∈ {$[0]_{12}$; $[2]_{12}$; $[6]_{12}$; $[8]_{12}$} has any immediate-odd-predecessor; such nodes are called 'inactive-nodes'.

(5) every odd-node x ∈ $[3]_6$ ≔ {$[3]_{12}$; $[9]_{12}$};

that is, x being an odd-multiple-of-3; is an 'odd-leaf-node';

none of its even predecessors has any immediate-odd-predecessor.

The system arborescence C represents the transitive-closure of the subset {1,5,3}; that is, [{1,5,3}C+]; under the inverse mapping of CTUHSK(3x+1)/4 system. We will show that this is exactly the same as [{1,5,3}H+] which has already been shown to be the entire set of positive integers.

8. Observations on the Critical Difference between H and C

The difference between the System(3X±1)/4 H and the CTUHSK(3x+1)/4 system C; as highlighted in the tabular representation of the system state transitions; corresponds to only the critical-(decision)-node (4m-1) and its connecting links.

The critical-(decision)-node (4m-1) ∈ {$[3]_{12}$; $[7]_{12}$; $[11]_{12}$} is linked to the detour-node (12m-2) ∈ {$[10]_{12}$} through the detour-link {(12m-2)⇐(4m-1)} ∈ C in the CTUHSK(3x+1)/4 system; whereas it is linked to the node (12m-4) ∈ {$[8]_{12}$} in the System(3X±1)/4. We designate that node (12m-4) as the *shortcut-node* and that link {(12m-4)⇐(4m-1)} ∈ H as the *shortcut-link*.

Now we design (1) transformation (reduction) H2C from the System(3X±1)/4 H to the CTUHSK(3x+1)/4 system C; and also (2) transformation (reduction) C2H from the CTUHSK(3x+1)/4 system C to the System(3X±1)/4 H; passing through a sequence of intermediate transition systems T; defined by some sequence of pruning-&-joining operations on the system arborescence.

9. From System(3X±1)/4 to the CTUHSK(3x+1)/4 system

Perform the following pruning-&-joining operation on the System(3X±1)/4, H; recursively/iteratively, to transform it into the CTUHSK(3x+1)/4 system C.

(1) PRUNING OPERATION: Disconnect the critical-decision-node (4m-1) from the shortcut-node (12m-4); by pruning the shortcut-link {(12m-4)⇐(4m-1)} in System(3X±1)/4 arborescence H. This pruning operation yields a free-hanging subtree, with the critical-decision-node (4m-1) as its root; that is denoted by [(4m-1)T+] ⊂ T; that is, a branch of the transition system arborescence T.

(2) JOINING OPERATION: Connect this critical-decision-node (4m-1) onto the detour-node (12m-2); which is known to be a hitherto inactive-node in System(3X±1)/4; by a joining operation; through a newly created detour-link {(12m-2)⇐(4m-1)}. Note that the detour-node (12m-2) is intended to act as an even-predecessor of an odd node (6m-1) in the prospective CTUHSK(3x+1)/4 system C being generated.

Thus, the entire subtree $[(4m-1)T+] \subset T$ that was pruned in pruning operation, now gets joined to the detour-node $(12m-2) \in [(6m-1)T+]$ as a new branch; where T is an intermediate transition system; consisting of the newly generated detour-links $\{(12m-2) \Leftarrow (4m-1)\}$ meant to be belonging to system C that is being generated; along with the remaining links of system H.

The above-described pruning-&-joining operation can be performed iteratively; that is, by replacing every shortcut-link $\{(12m-4) \Leftarrow (4m-1)\} \in H$ with the corresponding detour-link $\{(12m-2) \Leftarrow (4m-1)\} \in C$; for $m \in \{1, 2, 3, \text{etc.}\}$; without necessarily considering any network-based recursive algorithm to process the subtrees etc.; although a network-based algorithm may facilitate in a better conceptual understanding of the process involved in the pruning-&-joining operations therein.

Assertion-H2C-1:
The transformation (reduction) H2C does not affect the set of nodes in the system arborescence H; but replaces every shortcut-link by the corresponding detour-link.

This is clear from the above description of the pruning-&-joining operations used in the transformation H2C; neither any nodes are added nor any nodes are removed.

Assertion-H2C-2:
The transformation (reduction) H2C does not affect the connectedness property of the system; and specifically, the binary(inward)arborescence structure of the system is preserved.

The connectedness property of the system is unaffected by the exchange of shortcut-links for the detour-links, because pruning-& joining operation preserves

the inward arborescence structure. Specifically, the binary(inward)arborescence structure is preserved because, in every iteration, the pruned subtree [(4m-1)T+] ⊂ T is joined onto an already existing and hitherto inactive detour-node (12m-2) of the System(3X±1)/4 H; or of the intermediate transition system T.

Assertion-H2C-3:

The transitive-closure of the subset {1,5,3}; that is, [{1,5,3}H+]; under the inverse mapping of the original System(3X±1)/4 H; is contained in the transitive-closure of the subset {1,5,3}; under the inverse mapping of the resultant CTUHSK(3x+1)/4 system C; that is, {nodes(H)} = [{1,5,3}H+] ⊆ [{1,5,3}C+].

Assertiob-H2C-1 & Assertiob-H2C-2 together implies Assertion-H2C-3. The entire transformation process H2C; consists of a sequence of pruning-&-joining operations; wherein every pruning-&-joining operation preserves the original set of nodes in H while replacing every shortcut-link in H by the corresponding detour-link in C. Neither any nodes are removed nor any nodes are added during the pruning-&-joining operations. The entire set of nodes originally in H are now interlinked by the successor predecessor relation defined by C; whereas we must in general allow [{1,5,3}C+] to be possibly a superset of [{1,5,3}H+].

10. From the CTUHSK(3x+1)/4 system to System(3X±1)/4

A similar pruning-&-joining process can be described to transform CTUHSK(3x+1)/4 system C to System(3X±1)/4 H; that is, by replacing every

detour-link of the form {(12m-2)⇐(4m-1)} in C by the corresponding newly created shortcut-link of the form {(12m-4)⇐(4m-1)} in H; as described below:

(a) PRUNING OPERATION: Disconnect the critical-decision-node (4m-1) from the detour-node (12m-2); by pruning the detour-link {(12m-2)⇐(4m-1)} in CTUHSK(3x+1)/4 system arborescence C. This pruning operation yields a free-hanging subtree, with the critical-decision-node (4m-1) as its root; denoted by [(4m-1)T+] ⊂ T; that is, a branch of the transition system arborescence T.

(b) JOINING OPERATION: Connect this critical-decision-node (4m-1) onto the shortcut-node (12m-4); which is known to be an otherwise inactive-node in CTUHSK(3x+1)/4 system; by a joining operation; through a newly created shortcut-link {(12m-4)⇐(4m-1)} in H. Note that the shortcut-node (12m-4) is intended to act as an even-predecessor of a node (3m-1) in the prospective System(3X±1)/4 H being generated.

Thus, the entire subtree [(4m-1)T+] ⊂ T that was pruned in pruning operation, now gets joined to the shortcut-node (12m-4) ∈ [(3m-1)T+] as a new branch; where T is an intermediate transition-system; consisting of the newly generated shortcut-links {(12m-4)⇐(4m-1)} meant to be belonging to system H that is being generated, along with the remaining links of system C.

The above-described pruning-&-joining operation can be performed iteratively; that is, by replacing every detour-link {(12m-2)⇐(4m-1)} ∈ C with the corresponding shortcut-link {(12m-4)⇐(4m-1)} ∈ H; for m ∈ {1, 2, 3, etc.}; without necessarily considering any network-based recursive algorithm to process the subtrees etc.;

although a network-based algorithm may facilitate in a better conceptual understanding of the process involved in the pruning-&-joining operations therein.

Assertion-C2H-1:

The transformation (reduction) C2H does not affect the set of nodes in the system arborescence; but only replaces every detour-link by the corresponding shortcut-link.

This is clear from the above description of the pruning-&-joining operations used in the transformation C2H.

Assertion-C2H-2:

The transformation (reduction) C2H does not affect the connectedness property of the system; and specifically, the binary(inward)arborescence structure of the system is preserved.

The connectedness property of the system is unaffected by the exchange of shortcut-links for the detour-links, because pruning-& joining operation preserves the inward arborescence structure. Specifically, the binary(inward)arborescence structure is preserved because, in every iteration, the pruned subtree [(4m-1)T+] $\subset$ T is joined onto an already existing and hitherto inactive shortcut-node (12m-4) of the CTUHSK(3x+1)/4 system C; or of the intermediate transition system T.

Assertion-C2H-3:

The transitive-closure of the subset {1,5,3}; that is, [{1,5,3}C+]; under the inverse mapping of the original CTUHSK(3x+1)/4 system C; is contained in the transitive-

closure of the subset {1,5,3}; under the inverse mapping of the resultant System(3X±1)/4 H; that is, [{1,5,3}C+] ⊆ [{1,5,3}H+] = {nodes(H)}.

Assertiob-C2H-1 & Assertiob-C2H-2 together implies Assertion-C2H-3. The entire transformation process C2H; consists of a sequence of pruning-&-joining operations; wherein every pruning-&-joining operation preserves the original set of nodes in C while replacing every detour-link in C by the corresponding shortcut-link in H. Neither any nodes are removed nor any nodes are added during the pruning-&-joining operations. The entire set of nodes originally in C are now interlinked by the successor predecessor relation defined by H; whereas we must in general allow [{1,5,3}H+] to be a superset of [{1,5,3}C+].

11. Convergence of the CTUHSK(3x+1)/4 system

The system structure has been shown to be a binary(inward)arborescence with the trivial-cycle as the root; and every odd multiple of 3 being an 'odd-leaf-node'. Assertion-H-1 establishes the convergence of System(3X±1)/4 H to its trivial-cycle {(1⇐(2)⇐4)}. Assertion-H-2 establishes that the transitive-closure of the subset {1,5,3}; that is, [{1,5,3}H+]; under the inverse mapping of System(3X±1)/4; is the entire set of positive integers. Assertion-C-1 establishes that the CTUHSK(3x+1)/4 system C is exactly identical to the Collatz (3x+1) System. Assertion-H2C-1 & Assertion-C2H-1 together establish that the transformation process does not add or remove any nodes, but only causes an exchange between every shortcut-link in H and its corresponding detour-link in C. Assertion-H2C-2 & Assertion-C2H-2 together establish that the transformation process does not affect the connectedness property as well as the binary(inward)arborescence

structure of either of these systems. Assertion-H2C-3 & Assertion-C2H-3 together establish that the transitive-closure of the subset {1,5,3}; under the inverse mapping of one system; as represented by its binary(inward)arborescence; is exactly the same as that obtained for the other system; that is, [{1,5,3}C+] = [{1,5,3}H+] = {nodes(H)} = N. Therefore, these assertions together provide a definitive proof for the convergence of the CTUHSK(3x+1)/4 system to its trivial-cycle {(1⇐2⇐4)}. Because the CTUHSK(3x+1)/4 system, by its very definition, is simply a re-written re-formulation of the Collatz (3x+1) system, it provides a definitive proof for the Collatz Conjecture.

The above transformation process between System(3X±1)/4 H and the CTUHSK(3x+1)/4 system C; guarantees (through a domain-exhaustion argument); the non-existence of any open chains/loops or open (infinite) threads. The study of non-standard extraneous (rogue) objects (of whatever kind) is inconsequential to the present proof, because the Collatz Conjecture is stated strictly within the domain of natural numbers.

Note that every pruning-&-joining operation can be considered to be analogous to shifting from a shortcut sub-route to a detour sub-route or from a detour sub-route to a shortcut sub-route; wherein the shifting decision itself is being taken at every critical-(decision)-node; and wherein the set of nodes remain unchanged.

12. Summary

System(3X±1)/4 has been defined, very similar to the Collatz (3x+1) system; with slightly modified operations. We have also defined the CTUHSK(3x+1)/4 system,

and shown that to be exactly the same as the Collatz (3x+1) system. The convergence of System(3X±1)/4 to its trivial-cycle {(1⇐(2)⇐4)} is established using the persistent local 'odd-to-odd monotonicity' property of its defining function. A reversible (invertible) transformation between System(3X±1)/4 and CTUHSK(3x+1)/4 system has been designed, based on a recursive/iterative process consisting of pruning-&-joining operations performed on the binary(inward)arborescence structure of the system. This pruning-&-joining operation does not affect the binary(inward)arborescence structure and the connectedness property of either of these systems; because the pruned subtree is joined onto an already existing inactive-node of the system arborescence; without adding or removing any nodes. The transitive-closure of the subset {1,5,3} under the inverse mapping of one system is exactly the same as the transitive-closure of that subset under the inverse mapping of the other system; which establishes the convergence of the CTUHSK(3x+1)/4 system C to its trivial-cycle {(1⇐2⇐4)}; and therefore proving the Collatz Conjecture.

The Halemane-Conjecture on the Collatz (3x+1) sequence states that the maximum limit on number of odd (3x+1) operations required to reach its trivial-cycle {(1⇐2⇐4)}; starting from any given positive integer and moving along the Collatz sequence; is the given number itself; with the triad {(31⇐41⇐27)} as an exceptional limiting case.

13. Recommended Reading

[§1]. Wikipedia Page – https://en.wikipedia.org/wiki/Collatz_conjecture

[§2]. Jeffrey C Lagarias;
"The 3x+1 problem: An annotated bibliography (1963--1999)
(sorted by author)";
https://arxiv.org/abs/math/0309224

[§3]. Jeffrey C Lagarias;
"The 3x+1 Problem: An Annotated Bibliography, II (2000-2009)";
https://arxiv.org/abs/math/0608208

[§4]. Jeffrey C Lagarias;
"The 3x + 1 Problem: An Overview"
https://arxiv.org/abs/2111.02635

[§5]. Jeffrey C Lagarias;
"The Ultimate Challenge: The 3x+1 Problem";
American Mathematical Society; 2010.

[§6]. Richard K Guy;
"Don't Try to Solve These Problems";
The American Mathematical Monthly: Vol. 90, No. 1, pp. 35-41; 1983.
https://doi.org/10.1080/00029890.1983.11971148

14. Acknowledgment

The author wishes to acknowledge with gratitude the excellent openminded understanding & appreciation of the work by the Editorial Board & the Reviewers

Team, following the most basic principle of paying credit wherever-&-whenever it is rightfully due, without any misplacement. Note that the Collatz Conjecture has been referred to as the Collatz-Thwaites-Ulam-Hasse-Syracuse-Kakutani (CTUHSK) Conjecture and the CTUHSK(3x+1)/4 system has been named so to acknowledge all the known original sources thereof. Similarly, this author wishes that the System(3X±1)/4 be henceforth referred to as Halemane(3X±1)/4System; "not waiting for the Pope, for baptizing my brain-child" as he said elsewhere.

15. A Confession and An Assertion

I must necessarily confess that the *core idea behind this proof of the Collatz Conjecture is so stunningly and elusively simple*, that one may simply be taken aback in a profound wonder-struck jaw-drop-silence, possibly with an after-thought: "OMG, *how could it be that it never flashed on me any time earlier*"! However, I assert that क्लिष्टविशयसरलीकरणमेव पाण्डित्यलक्षणम् | न तु सरलविषयसंक्लिष्टीकरणम् || That is, *to make the complex simple is a mark of mastery*; *not to make the simple complex.*

16. Declaration Regarding Affiliation and Funding

I, Dr(Prof) Keshava Prasad Halemane, hereby declare that I am a Professor retired as on 2017JAN31 from National Institute of Technology Karnataka Surathkal India, and I am not affiliated to any institution or organization or corporation or any other agency or whatever. This research work has been conducted entirely by me on my own as an Independent Researcher, and that I have not received any funding from any source other than my own savings, and I do not have any obligations or encumbrances of any kind, neither financial nor legal nor of any other

kind, regarding the contents of the manuscript - of which I am the original author and creator. Also, I hereby declare there is no conflict of interests.

17. Author Rights Retention Notice

The author as the original creator of this work asserts exclusive ownership and retains full-&-complete Intellectual Property Rights (IPR) and CopyRight for this (as well as all its previous versions, including all the versions archived in various repositories, all of them with CC-BY-NC-ND 4.0 International Public Copyright License) and any/every Author Accepted Manuscript (AAM) version arising from this manuscript) in perpetuity, for the maximum duration permitted by law. For the purpose of publication, the author hereby grants a non-exclusive publishing rights to the publisher. The author's IPR including all commercial and derivative rights are retained in full as the exclusive intellectual property of the author and shall pass to his designated legal heirs (the three children mentioned herein below) in perpetuity, for the maximum duration permitted by law.

18. Dedication

To my **ಅಜ್ಜ**(ajja) Karinja Halemane Keshava Bhat & **ಅಜ್ಜಿ**(ajji) Thirumaleshwari, **ಅಪ್ಪ**(appa) Shama Bhat & **ಅಮ್ಮ**(amma) Thirumaleshwari, for their *teachings through love*, that *quality matters more than quantity*; to my wife Vijayalakshmi for her *ever consistent love & support*; to my daughter Sriwidya.Bharati and my twin sons Sriwidya.Ramana & Sriwidya.Prawina for their *love & affection*.